

ARQILA SURYA PUTRA

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Bandung, Indonesia

PROFESSIONAL SUMMARY

Production AI Engineer building realtime LLM and vision pipelines that survive real world constraints. Operates moderation for 2,800 users and edge YOLO at 94% accuracy in 120ms on Raspberry Pi, plus grounded RAG with tool calling. Ensures reliability through idempotent webhooks, retry queues, and gating AI decisions before they reach users.

EDUCATION

Institut Teknologi Bandung (ITB)

Bachelor of Engineering, Information and System Technology

2023 - 2027

Bandung, Indonesia

EXPERIENCE

Containder Indonesia

Apr 2025 - Aug 2025

AI Engineer Intern / Remote

- Built YOLOv5 and OpenCV prototype on Raspberry Pi for bottle detection and volume estimation, achieving 94% accuracy with 6.7x faster inference (800ms to 120ms) with optimized real-time inference preprocessing.
- Evaluated 2 architectures (ArUco markers vs. YOLO) across latency, accuracy, and hardware cost to select YOLO for production, enabling <150ms physical feedback in IoT smart bins.
- Deployed FastAPI + WebSocket backend for real-time edge-to-cloud messaging, supporting 10+ concurrent bins with <50ms round-trip and continuous operation.

Ganesa Space

Sep 2024 - Present

Founder and Full-Stack Developer / Bandung, West Java

- Scaled platform to 2,800+ active users and 269,000+ events in 3 months by architecting PostgreSQL with Row-Level Security, indexing, and Supabase migrations, supporting 60,000+ page views and 2M+ weekly X impressions.
- Automated up to 100 daily menfess deliveries via cron and webhook processing, IFTTT pipelines and async job queues with retry logic, eliminating ~14 hrs/week manual posting and achieving 99% on-time delivery.
- Integrated Xendit gateway processing 170+ monthly transactions with idempotent webhooks, state machines, and signature verification, achieving 99.5% reconciliation with zero duplicate charges.
- Cut deploy time to <5 mins by managing VPS via Coolify, Docker, and GitHub Actions CI/CD maintaining reliable releases for 2,800+ users.

Pakuwon Group

Jun 2026 - Aug 2026

AI Engineer Intern / South Jakarta, DKI Jakarta

- Built Vehicle Compliance Tracker for 114+ vehicles, automating daily morning email (Resend) + WhatsApp (Fonnte) reminders with idempotent logs, replacing ~10 hrs/week manual Excel tracking and reducing missed renewals to zero.
- Engineered Next.js 16 + Supabase + Clerk (restricted provisioning) + Cloudflare R2 for versioned compliance docs, securing cron endpoints with bearer tokens and hourly sweeps clearing 100% of abandoned uploads.
- Implemented bulk data pipeline with duplicate detection and date-anomaly validation, preview and audit logging, migrating production fleet data from master Excel into normalized relational schema.

PROJECTS

Taleka

Feb 2026 - May 2026

- Launched mobile-first Semai preservation platform with Elder Studio voice archiving and grounded AI Coach, combining dictionary-constrained translation, OmniASR transcription, and Supabase in Ionic/React/Capacitor, **winning Champion at BorNEO HackWknd 2026 among 103 teams from 7 ASEAN countries.**

AquaMine AI

Jan 2026 - Feb 2026

- Engineered IoT early warning platform for acid mine drainage cutting 48-hour lab confirmation to real-time detection, combining MQTT/TimescaleDB telemetry, TimeGPT forecasting, and drone YOLOv8 verification of Yellow Boy precipitate with geospatial heatmap, **winning 2nd Runner Up at ISMC XV.**

Kirana

Mar 2026 - Apr 2026

- Developed full-stack dyslexia screening prototype with Smart Reader adaptive PDF, combining YOLO handwriting, faster-whisper transcription, and WebGazer eye tracking via Next.js/FastAPI and LangGraph reasoning, **winning 3rd Place at HackFest UC 2026.**

TECHNICAL SKILLS

Programming Languages: Javascript, TypeScript, Python, SQL, Go
Frontend and Mobile: Next.js, React, Ionic, Capacitor, Tailwind CSS, Nativewind
Backend and Cloud: FastAPI, Node.js, PostgreSQL, Supabase, MongoDB, TimescaleDB, Redis, WebSockets, MQTT
AI and ML: RAG, LangChain, LangGraph, Vector Search, Groq, YOLOv5/v8, OpenCV, Whisper, OmniASR, WebGazer, TimeGPT, Function Calling, LLM, OpenRouter
Infrastructure: Docker, GitHub Actions, Coolify, Vercel, Cloudflare R2, Dockify, Greptile, CodeRabbit, Blacksmith

AWARDS

Best Innovation Award International Leadership in Action Summit <i>Salman ITB</i>	2024
1st Runner Up COMPFEST 17 AI Innovation Challenge <i>Universitas Indonesia</i>	2025
Ganesha Karya Awardee <i>Institut Teknologi Bandung</i>	2025
Champion at BorNEO HackWknd Hackathon 2026 <i>Universiti Malaysia Sarawak and Sarawak Digital Economy Corporation</i>	2026
2nd Runner Up ISMC XV Hackathon <i>Indonesia Student Mining Competition</i>	2026
2nd Runner Up NUS FoodTech Competition <i>National University of Singapore</i>	2026
3rd Place HackFest UC Hackathon 2026 <i>Univesitas Ciputra</i>	2026